

Software Requirements Specification

(SRS Document)

Plagiarism Checking Portal

A Web-Based System for Automated Detection of Textual Plagiarism

Submitted by: Tathagata Chakraborty
Roll Number: UG/SOET/30/24/144
Section: G
Programme: B.Tech (Computer Science and Engineering)

Submitted in partial fulfilment of the requirements
for the academic project / minor project evaluation
of the B.Tech CSE programme

August 31, 2026

Contents

1. Introduction	3
0.1 Purpose	3
0.2 Scope	3
0.3 Definitions, Acronyms and Abbreviations	4
0.4 References	4
0.5 Overview of the Document	4
2. Overall Description	6
0.6 Product Perspective	6
0.7 Product Functions	6
0.8 User Classes and Characteristics	7
0.9 Operating Environment	7
0.10 Design and Implementation Constraints	7
0.11 Assumptions and Dependencies	8
3. System Features and Functional Requirements	9
0.12 User Registration and Login	9
0.13 User Authentication and Session Management	10
0.14 Document/File Upload	11
0.15 Text Extraction from Uploaded Documents	11
0.16 Plagiarism Detection	12
0.17 Similarity Percentage Calculation	13
0.18 Comparison with Existing Documents/Content	13
0.19 Highlighting/Identifying Matching Content	14
0.20 Generation of Plagiarism Reports	14
0.21 Report Viewing	15
0.22 Report Download	15
0.23 Document History	16
0.24 User Dashboard	16
0.25 Admin Functionality	17
0.26 Database Management	17
0.27 Error Handling	18
4. External Interface Requirements	19
0.28 User Interface	19
0.29 Hardware Interfaces	19
0.30 Software Interfaces	19
0.31 Communication Interfaces	19

5. Non-Functional Requirements	20
6. System Models	21
0.32 Use Case Diagram	21
0.33 Data Flow Diagram	22
0.34 System Architecture	23
0.35 Entity Relationship Diagram	23
0.36 Activity Diagram	23
0.37 Sequence Diagram	24
7. Database Requirements	26
0.38 Users Table	26
0.39 Documents Table	26
0.40 Plagiarism Reports Table	27
0.41 Similarity Results Table	27
0.42 Admin Table	28
0.43 Document History Table	28
8. System Constraints and Limitations	29
9. Future Enhancements	30
10. Acceptance Criteria	31
11. Conclusion	32
Appendix	33
0.44 Sample Plagiarism Report Structure	33
0.45 Glossary	33
0.46 Requirement Traceability Matrix	34

1. Introduction

0.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed and structured description of the **Plagiarism Checking Portal**, a web-based application intended to help students, faculty, and academic institutions detect textual plagiarism in submitted documents. This document describes the functional and non-functional requirements, system models, database design, and constraints of the proposed system so that developers, evaluators, and stakeholders share a common and unambiguous understanding of what the system is expected to do.

This SRS is prepared as part of the academic project submission for the B.Tech CSE programme and is intended to serve both as a design reference during development and as an evaluation document for academic assessment.

0.2 Scope

The Plagiarism Checking Portal is a web application that allows registered users (primarily students and faculty) to upload textual documents and receive an automated report indicating the extent to which the submitted content overlaps with existing reference material. The system will:

- Allow users to register, log in, and manage their profile securely.
- Allow users to upload documents in common formats such as `.txt`, `.pdf`, and `.docx`.
- Extract text content from the uploaded documents.
- Compare the extracted text against a reference corpus (previously submitted documents and, optionally, external sources) using a similarity detection module.
- Generate a plagiarism report showing an overall similarity percentage and highlighting the matched sections.
- Allow users to view, download, and maintain a history of their previous reports.
- Provide an administrative interface for managing users, documents, and system-level configuration.

The system is not intended to replace human academic judgement; rather, it is designed as a supporting tool that assists in the identification of potentially copied content. Detection of plagiarism in non-textual media (images, audio, source code) is outside the scope of the current version, although such enhancements are discussed in Chapter 9.

0.3 Definitions, Acronyms and Abbreviations

Term	Description
SRS	Software Requirements Specification
FR	Functional Requirement
NFR	Non-Functional Requirement
UC	Use Case
UI	User Interface
API	Application Programming Interface
DB	Database
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
HTTPS	Hyper Text Transfer Protocol Secure
JWT	JSON Web Token (used for authentication sessions)
Corpus	The collection of reference documents against which a submission is compared
Similarity Score	A numerical (percentage) measure of textual overlap between a submitted document and the reference corpus

0.4 References

- [1] IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
- [2] Sommerville, I., *Software Engineering*, 10th Edition, Pearson Education.
- [3] Pressman, R.S., *Software Engineering: A Practitioner's Approach*, 8th Edition, McGraw-Hill.
- [4] Official documentation of Python/Node.js web frameworks used for reference during design (Flask/Django/Express).
- [5] Class lecture notes and project guidelines provided for the B.Tech CSE academic project.

0.5 Overview of the Document

The remainder of this document is organised as follows: Chapter 2 gives an overall description of the product, its functions, and the environment in which it operates. Chapter 3 lists detailed functional requirements for each system feature. Chapter 4 describes external interface requirements. Chapter 5 specifies non-functional requirements. Chapter 6 presents system models such as use case, data flow, architecture, ER, activity, and sequence diagrams. Chapter 7 details the database design. Chapters 8 and 9 discuss constraints and future enhancements respectively. Chapter 10 lists acceptance criteria,

Chapter 11 concludes the document, and the Appendix provides supplementary material including a glossary and a requirement traceability matrix.

2. Overall Description

0.6 Product Perspective

The Plagiarism Checking Portal is designed as a standalone web application with a client-server architecture. It is not dependent on any existing legacy system, although it may optionally integrate with an institution's Learning Management System (LMS) in future versions (see Chapter 9). The system consists of the following high-level components:

- A **front-end web interface** through which users interact with the system.
- A **back-end API/application server** that handles business logic, authentication, and orchestration of requests.
- A **text extraction module** that converts uploaded documents into plain text.
- A **similarity detection engine** that compares extracted text against the reference corpus and computes similarity scores.
- A **database** that stores user data, documents, reports, and history.

0.7 Product Functions

At a high level, the system provides the following major functions, each of which is elaborated in Chapter 3:

- User registration, login, and profile/session management.
- Secure upload of documents for plagiarism checking.
- Automatic extraction of text from supported file formats.
- Comparison of the extracted text with the existing document corpus.
- Computation of an overall similarity percentage along with section-wise matches.
- Generation of a structured plagiarism report.
- Viewing and downloading of reports.
- Maintaining a history of previously checked documents.
- A dashboard summarising a user's activity and past submissions.
- Administrative functions for managing users and monitoring system usage.

0.8 User Classes and Characteristics

User Class	Characteristics
Student User	Uploads assignments/reports for a self-check before final submission; expected to have basic computer literacy; uses the dashboard and report features most frequently.
Faculty User	Uploads student submissions to verify originality; may need to compare multiple documents against each other in addition to the general corpus; requires access to detailed similarity reports.
Administrator	Manages user accounts, monitors system health, maintains the reference corpus, and configures system-level settings such as similarity thresholds.
Guest/Unregistered Visitor	Can view basic information about the portal (landing page, login/registration) but cannot upload documents or view reports until registered.

0.9 Operating Environment

The system is designed to operate as follows:

- **Client side:** Any modern web browser (Google Chrome, Mozilla Firefox, Microsoft Edge) on Windows, macOS, Linux, or mobile devices with a responsive layout.
- **Server side:** A backend application server (e.g. built using a framework such as Django, Flask, or Node.js/Express) hosted on a cloud platform or institutional server.
- **Database server:** A relational database system (e.g. MySQL or PostgreSQL) used to persist structured data.
- **Network:** The system communicates over HTTPS to ensure secure transmission of documents and user credentials.

0.10 Design and Implementation Constraints

- The system must be implemented using technologies that support secure handling of user-uploaded files.
- The similarity detection module must be able to process reasonably sized text documents (a few thousand words) within an acceptable time frame.
- The system must comply with basic data protection principles, since it stores personal user data and potentially sensitive academic work.
- The initial version of the system will support a limited set of file formats (.txt, .pdf, .docx); other formats are considered future work.
- The project is being developed as part of an academic curriculum and is therefore constrained by the timeline and resources typically available for a B.Tech project.

0.11 Assumptions and Dependencies

- It is assumed that users will have consistent internet access while using the portal.
- It is assumed that uploaded documents are primarily in English; multilingual support is listed as a future enhancement.
- The accuracy of plagiarism detection depends on the size and quality of the reference corpus available to the system.
- The system depends on third-party libraries for text extraction (e.g. PDF/DOCX parsing libraries) and, optionally, external APIs for extended similarity checking against web sources.

3. System Features and Functional Requirements

This chapter describes each major feature of the Plagiarism Checking Portal in detail. For every feature, the description, preconditions, the main (successful) workflow, alternative/error workflows, and the associated functional requirements are provided. Each functional requirement is assigned a unique identifier of the form **FR-XX** for traceability purposes (see Appendix, Section 0.46).

0.12 User Registration and Login

Description: Allows a new user to create an account and existing users to log in to access the portal's features.

Preconditions: The user has a valid email address and internet access. For login, the user must already possess a registered account.

Main Workflow:

1. The user navigates to the registration page and enters details such as name, email, password, and role (student/faculty).
2. The system validates the input and checks whether the email is already registered.
3. On successful validation, the system creates a new user record and sends a confirmation notification.
4. The user logs in using registered email and password credentials.
5. The system validates the credentials and redirects the user to the dashboard.

Alternative/Error Workflow:

- If the email is already registered, the system displays an appropriate error message and prompts the user to log in instead.
- If the password does not meet the minimum security requirements, the system rejects the registration and displays guidance.
- If login credentials are incorrect, the system displays an authentication failure message without revealing whether the email or password was incorrect (for security reasons).

ID	Requirement
FR-01	The system shall allow a new user to register by providing name, email address, password, and role.
FR-02	The system shall validate that the email address is unique before creating an account.
FR-03	The system shall enforce a minimum password complexity policy during registration.
FR-04	The system shall allow a registered user to log in using email and password.
FR-05	The system shall lock an account temporarily after a configurable number of consecutive failed login attempts.

0.13 User Authentication and Session Management

Description: Ensures that only verified users can access protected areas of the portal and that user sessions are handled securely.

Preconditions: The user has successfully registered.

Main Workflow:

1. Upon successful login, the system generates a secure session token (e.g. a JWT) for the user.
2. The token is used to authorise subsequent requests to protected endpoints (upload, dashboard, reports).
3. The session remains valid until the user logs out or the token expires.

Alternative/Error Workflow:

- If the session token is invalid or has expired, the system redirects the user to the login page.
- If a user attempts to access an admin-only resource without sufficient privileges, the system denies access with a "403 Forbidden" style response.

ID	Requirement
FR-06	The system shall issue a secure session token upon successful authentication.
FR-07	The system shall verify the session token on every request to a protected resource.
FR-08	The system shall automatically expire sessions after a configurable period of inactivity.
FR-09	The system shall restrict access to administrative functionality based on user role.

0.14 Document/File Upload

Description: Allows a logged-in user to upload one or more documents for plagiarism checking.

Preconditions: The user is authenticated and has a document in a supported format.

Main Workflow:

1. The user selects a file from their device and submits it through the upload form.
2. The system verifies the file type and size against configured limits.
3. The file is stored securely on the server, and a corresponding database record is created.
4. The system queues the document for text extraction and plagiarism checking.

Alternative/Error Workflow:

- If the file format is unsupported, the system rejects the upload and informs the user of accepted formats.
- If the file exceeds the maximum allowed size, the system displays an appropriate error message.
- If the upload is interrupted, the system discards any partially uploaded file.

ID	Requirement
FR-10	The system shall allow authenticated users to upload documents in .txt, .pdf, or .docx format.
FR-11	The system shall reject files that exceed a configured maximum file size.
FR-12	The system shall store uploaded documents in a secure, access-controlled storage location.
FR-13	The system shall associate each uploaded document with the uploading user's account.

0.15 Text Extraction from Uploaded Documents

Description: Converts the content of an uploaded document into plain text suitable for comparison by the similarity detection engine.

Preconditions: A document has been successfully uploaded and stored.

Main Workflow:

1. The system identifies the file type of the uploaded document.
2. An appropriate parser (plain text reader, PDF parser, or DOCX parser) extracts the textual content.
3. Extracted text is normalised (e.g. whitespace and encoding cleanup) and stored temporarily for processing.

Alternative/Error Workflow:

- If the document is corrupted or unreadable, the system marks the document status as "extraction failed" and notifies the user.
- If the document contains no extractable text (e.g. a scanned image with no OCR support), the system informs the user that the document could not be processed.

ID	Requirement
FR-14	The system shall automatically extract text content from supported document formats after upload.
FR-15	The system shall normalise extracted text (whitespace, encoding) before comparison.
FR-16	The system shall mark a document's status as failed if text extraction is unsuccessful.

0.16 Plagiarism Detection

Description: Compares the extracted text of a submitted document against a reference corpus to identify overlapping content.

Preconditions: Text has been successfully extracted from the uploaded document.

Main Workflow:

1. The system breaks the extracted text into comparable units (e.g. sentences or n-gram segments).
2. Each unit is compared against corresponding units in the reference corpus using a similarity detection algorithm.
3. Matching segments above a configured similarity threshold are recorded along with their source references.

Alternative/Error Workflow:

- If the reference corpus is empty or unavailable, the system informs the user that comparison could not be completed and logs the issue for the administrator.
- If the comparison process times out for very large documents, the system notifies the user and suggests splitting the document.

ID	Requirement
FR-17	The system shall compare extracted text against the reference document corpus using a similarity detection algorithm.
FR-18	The system shall identify matching segments that exceed a configurable similarity threshold.
FR-19	The system shall record the source of each matched segment for reporting purposes.

ID	Requirement
FR-20	The system shall notify the user if plagiarism detection cannot be completed due to a system or data error.

0.17 Similarity Percentage Calculation

Description: Computes an overall similarity score representing the proportion of the submitted document that matches existing content.

Preconditions: Plagiarism detection has completed for the document.

Main Workflow:

1. The system aggregates the length of all matched segments identified during detection.
2. The system calculates the similarity percentage as the ratio of matched content to total document length.
3. The overall similarity percentage is stored along with the document's report record.

Alternative/Error Workflow:

- If the document length cannot be determined (e.g. due to extraction issues), the system reports the similarity score as "unavailable" rather than showing an incorrect value.

ID	Requirement
FR-21	The system shall calculate an overall similarity percentage for each submitted document.
FR-22	The system shall express the similarity percentage as a value between 0% and 100%.
FR-23	The system shall store the computed similarity percentage with the corresponding report.

0.18 Comparison with Existing Documents/Content

Description: Allows the system to check a newly submitted document not only against a general corpus but also against specific previously submitted documents (e.g. for faculty checking multiple student submissions against each other).

Preconditions: At least one reference document exists in the corpus or a specific comparison set has been selected.

Main Workflow:

1. The user (typically faculty) selects a set of documents to compare against one another, or the system automatically uses the full corpus.
2. The system performs pairwise or corpus-wide comparison as configured.
3. Results are compiled into a report showing similarity between the selected documents.

Alternative/Error Workflow:

- If fewer than two documents are selected for pairwise comparison, the system prompts the user to select an additional document.

ID	Requirement
FR-24	The system shall allow comparison of a submitted document against the general reference corpus.
FR-25	The system shall allow faculty users to perform pairwise comparison between selected documents.

0.19 Highlighting/Identifying Matching Content

Description: Visually indicates which portions of the submitted document matched existing content, to help the user quickly identify problem areas.

Preconditions: Plagiarism detection has been completed and matched segments have been identified.

Main Workflow:

1. The system marks matched segments within the document text (e.g. using highlighted colour or underline in the report view).
2. Each highlighted segment is linked to its corresponding source in the reference corpus.

Alternative/Error Workflow:

- If no matches are found, the system displays the document as fully original with no highlights.

ID	Requirement
FR-26	The system shall visually highlight matched segments within the report view of a document.
FR-27	The system shall provide a reference/link to the source of each highlighted match where available.

0.20 Generation of Plagiarism Reports

Description: Compiles the results of the plagiarism check into a structured report.

Preconditions: Similarity calculation and match highlighting have been completed for the document.

Main Workflow:

1. The system generates a report containing the document title, submission date, overall similarity percentage, and a list of matched segments with sources.
2. The report is saved and linked to the user's account and the corresponding document.

Alternative/Error Workflow:

- If report generation fails due to a system error, the document status is set to "report generation failed" and the user is notified.

ID	Requirement
FR-28	The system shall automatically generate a structured plagiarism report after detection is complete.
FR-29	The report shall include the overall similarity percentage, matched segments, and their sources.

0.21 Report Viewing

Description: Allows users to view previously generated plagiarism reports through the web interface.

Preconditions: At least one report has been generated for the user's account.

Main Workflow:

1. The user navigates to the reports section of the dashboard.
2. The user selects a specific report to view in detail, including highlighted matches.

Alternative/Error Workflow:

- If the requested report does not exist or does not belong to the user, the system denies access and displays an error message.

ID	Requirement
FR-30	The system shall allow a user to view the full details of any report associated with their account.
FR-31	The system shall prevent users from viewing reports belonging to other users, except administrators.

0.22 Report Download

Description: Allows a user to download a copy of the plagiarism report (e.g. as a PDF) for offline reference or submission.

Preconditions: A report has been generated and is available for the user.

Main Workflow:

1. The user selects the "Download Report" option for a given document.
2. The system generates or retrieves a downloadable file (e.g. PDF) of the report.

Alternative/Error Workflow:

- If the download file cannot be generated, the system displays an error and allows the user to retry.

ID	Requirement
FR-32	The system shall allow users to download their plagiarism reports in a common file format such as PDF.

0.23 Document History

Description: Maintains a record of all documents previously submitted by a user along with their associated reports.

Preconditions: The user has submitted at least one document in the past.

Main Workflow:

1. The user accesses the "History" section from the dashboard.
2. The system displays a chronological list of previously submitted documents with status and similarity percentage.

Alternative/Error Workflow:

- If no documents have been submitted, the system displays an appropriate empty-state message.

ID	Requirement
FR-33	The system shall maintain a history of all documents submitted by a user.
FR-34	The system shall display the submission date, status, and similarity percentage for each historical entry.

0.24 User Dashboard

Description: Provides a central summary view for a logged-in user, showing recent activity, quick access to upload, and an overview of report statistics.

Preconditions: The user is logged in.

Main Workflow:

1. Upon login, the user is redirected to the dashboard.
2. The dashboard displays recent uploads, pending checks, and quick links to reports and history.

Alternative/Error Workflow:

- If dashboard data cannot be retrieved due to a temporary system issue, a friendly error message is shown with an option to retry.

ID	Requirement
FR-35	The system shall present a dashboard summarising the user's recent uploads and reports.
FR-36	The dashboard shall provide direct navigation links to upload, history, and report sections.

0.25 Admin Functionality

Description: Provides administrative users with tools to manage user accounts, monitor system usage, and maintain the reference corpus.

Preconditions: The logged-in user has administrator privileges.

Main Workflow:

1. The administrator logs in and accesses the admin panel.
2. The administrator can view, activate, deactivate, or delete user accounts.
3. The administrator can view system-wide statistics and manage the reference corpus.

Alternative/Error Workflow:

- If a non-administrator attempts to access the admin panel, the system denies access.

ID	Requirement
FR-37	The system shall provide an administrative panel accessible only to users with admin privileges.
FR-38	The system shall allow administrators to manage (activate/deactivate/delete) user accounts.
FR-39	The system shall allow administrators to view system-wide usage statistics.

0.26 Database Management

Description: Ensures reliable storage, retrieval, and integrity of all data related to users, documents, and reports.

Preconditions: The database server is running and accessible to the application.

Main Workflow:

1. The application performs create, read, update, and delete (CRUD) operations on the database as required by user actions.
2. The database enforces referential integrity between related tables (e.g. a report cannot exist without a corresponding document).

Alternative/Error Workflow:

- If the database connection fails, the system displays a maintenance message and logs the failure for administrator review.

ID	Requirement
FR-40	The system shall persist all user, document, and report data in a relational database.
FR-41	The system shall maintain referential integrity between related database tables.

0.27 Error Handling

Description: Ensures that the system responds gracefully to unexpected conditions and provides meaningful feedback to the user.

Preconditions: None (applies globally across the system).

Main Workflow:

1. Whenever an operation fails, the system catches the error and logs relevant details for debugging.
2. The user is shown a clear, non-technical error message describing what went wrong and any suggested next step.

Alternative/Error Workflow:

- In the case of unhandled or unexpected exceptions, the system displays a generic error page rather than exposing internal details.

ID	Requirement
FR-42	The system shall log all errors with sufficient detail to support debugging.
FR-43	The system shall present user-friendly error messages without exposing internal technical details.

4. External Interface Requirements

0.28 User Interface

The system shall provide a responsive web-based user interface consisting of, at minimum, the following screens: landing/login page, registration page, dashboard, document upload page, report view page, history page, and (for administrators) an admin panel. The interface shall follow consistent navigation and visual styling across all pages, and shall be usable on both desktop and mobile browsers.

0.29 Hardware Interfaces

The system does not require any specialised hardware. It requires standard computing hardware capable of running a modern web browser on the client side and standard server hardware (physical or cloud-based virtual machine) capable of hosting the application and database on the server side.

0.30 Software Interfaces

- **Operating System:** The server may run on any operating system capable of hosting the chosen backend framework (e.g. Linux-based server).
- **Backend Framework:** A web application framework such as Django, Flask, or Node.js/Express for implementing the API and business logic.
- **Database:** A relational database management system such as MySQL or PostgreSQL.
- **Text Extraction Libraries:** Libraries capable of parsing .pdf and .docx files into plain text.
- **Similarity Detection Module:** A text similarity/NLP library or custom algorithm (e.g. based on n-gram comparison, cosine similarity of vectorised text, or fingerprinting techniques).

0.31 Communication Interfaces

The system shall use HTTPS for all communication between the client browser and the server to ensure that user credentials and uploaded documents are transmitted securely. Internal communication between application components (e.g. application server and database) shall occur over a secured internal network connection.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall complete plagiarism detection for a document of up to 5000 words within a reasonable processing time (target: under 60 seconds) under normal load.
NFR-02	Security	The system shall store user passwords using a secure hashing algorithm and never store passwords in plain text.
NFR-03	Security	The system shall enforce HTTPS for all client-server communication.
NFR-04	Reliability	The system shall be designed to recover gracefully from component failures without loss of previously stored data.
NFR-05	Availability	The system shall aim for an uptime suitable for regular academic use, with planned maintenance windows communicated in advance.
NFR-06	Usability	The system shall provide an intuitive interface requiring minimal training for a typical student or faculty user.
NFR-07	Scalability	The system architecture shall support an increasing number of registered users and documents without requiring a fundamental redesign.
NFR-08	Maintainability	The system shall be developed using modular components to simplify future maintenance and feature additions.
NFR-09	Portability	The web-based front end shall function correctly across major modern browsers without requiring browser-specific code.
NFR-10	Data Privacy	The system shall restrict access to uploaded documents and reports strictly to the owning user and authorised administrators.

6. System Models

This chapter presents the key system models used to describe the structure and behaviour of the Plagiarism Checking Portal. Where full graphical diagrams could not be embedded directly, simplified TikZ-based diagrams or structured descriptions are provided; these are sufficient to convey the intended design and can be replaced with more detailed diagrams (e.g. drawn in draw.io or Lucidchart) during implementation.

0.32 Use Case Diagram

The primary actors of the system are the **Student/Faculty User**, the **Administrator**, and, indirectly, the **Similarity Detection Engine**. Table 20 summarises the main use cases.

ID	Use Case	Primary Actor
UC-01	Register Account	Student/Faculty User
UC-02	Login	Student/Faculty User, Administrator
UC-03	Upload Document	Student/Faculty User
UC-04	View Report	Student/Faculty User
UC-05	Download Report	Student/Faculty User
UC-06	View Document History	Student/Faculty User
UC-07	Manage Users	Administrator
UC-08	Manage Reference Corpus	Administrator
UC-09	Compare Multiple Documents	Faculty User

Table 20: Summary of Use Cases

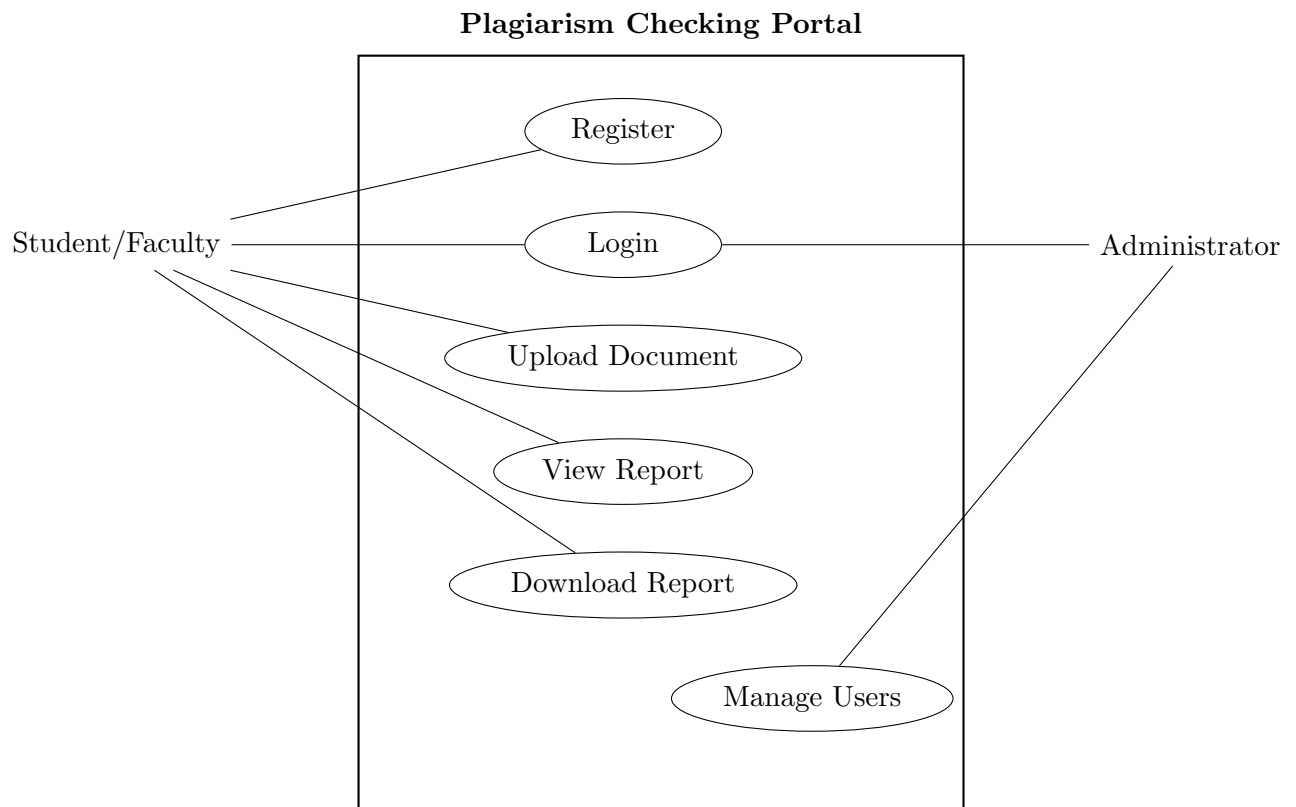


Figure 1: Simplified Use Case Diagram

0.33 Data Flow Diagram

Figure 2 shows a level-1 data flow diagram describing how a document moves through the system, from upload to report generation.

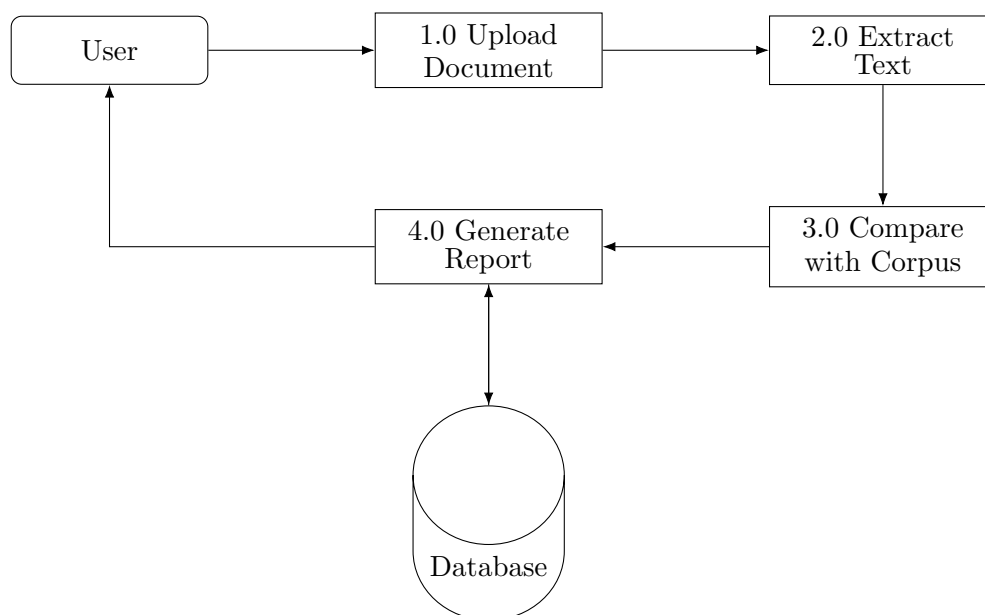


Figure 2: Level-1 Data Flow Diagram

0.34 System Architecture

The system follows a layered client-server architecture, as illustrated in Figure 3.

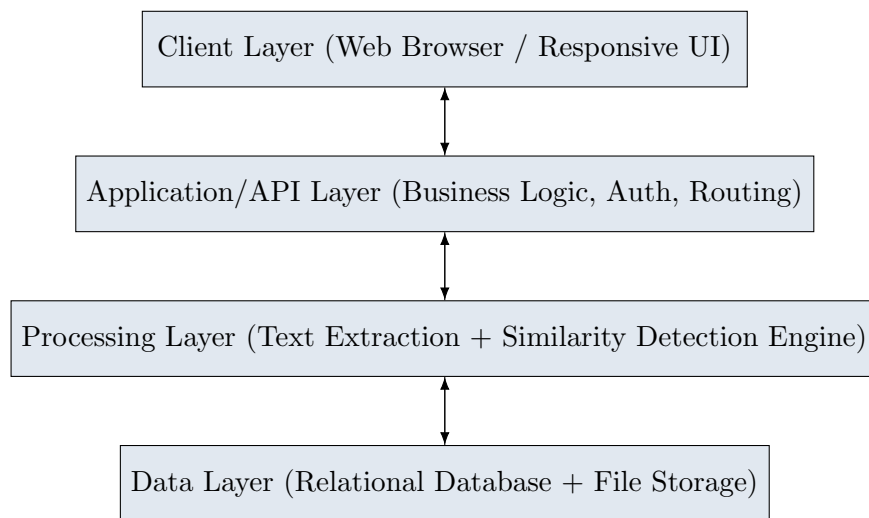


Figure 3: High-Level System Architecture

0.35 Entity Relationship Diagram

Figure 4 shows a simplified entity relationship diagram of the core database entities. A more detailed schema is presented in Chapter 7.

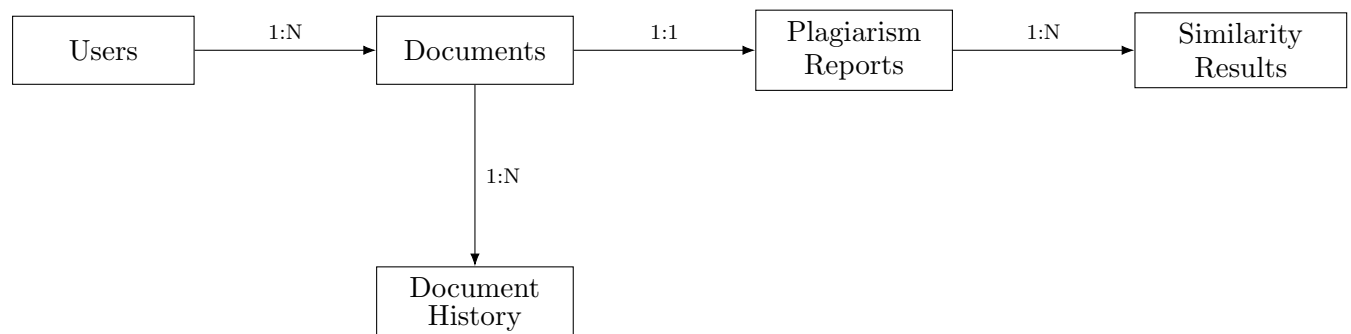


Figure 4: Simplified Entity Relationship Diagram

0.36 Activity Diagram

Figure 5 describes the overall activity flow for a document submission, from upload to report generation.

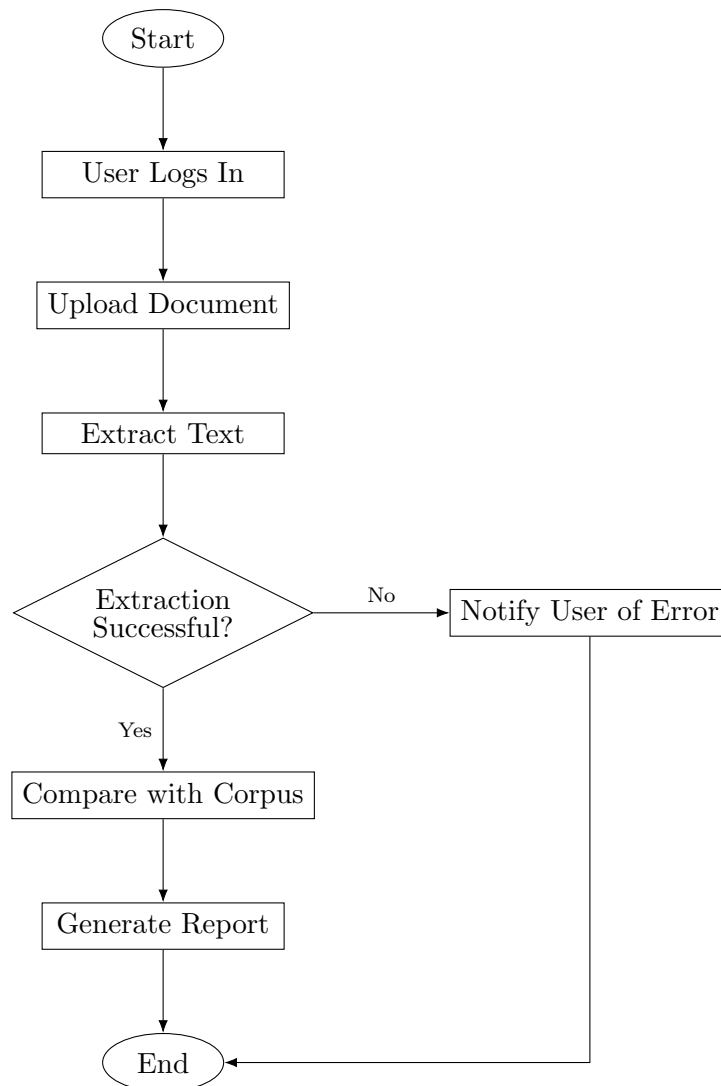


Figure 5: Activity Diagram for Document Submission

0.37 Sequence Diagram

Figure 6 illustrates the sequence of interactions between the user, application server, processing engine, and database during a typical plagiarism check.

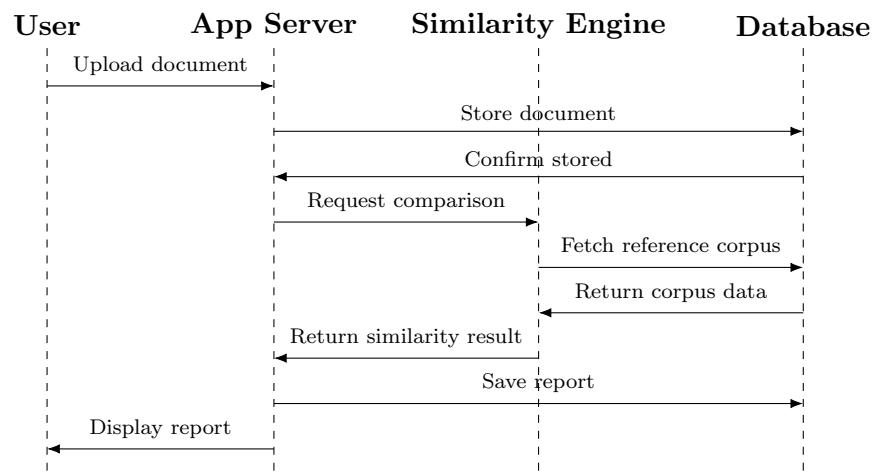


Figure 6: Sequence Diagram for Plagiarism Check Process

7. Database Requirements

This chapter describes the main database entities required to support the system’s functionality. A relational database model is assumed, with the entities described below linked through foreign key relationships as shown in the ER diagram (Figure 4).

0.38 Users Table

Attribute	Data Type	Description	Constraints
user_id	INT	Unique identifier for the user	Primary Key, Auto Increment
name	VARCHAR(100)	Full name of the user	NOT NULL
email	VARCHAR(100)	Login email address	NOT NULL, UNIQUE
password_hash	VARCHAR(255)	Hashed password	NOT NULL
role	VARCHAR(20)	Student / Faculty / Admin	NOT NULL
created_at	DATETIME	Account creation timestamp	NOT NULL, Default: current time

0.39 Documents Table

Attribute	Data Type	Description	Constraints
document_id	INT	Unique identifier for the document	Primary Key, Auto Increment
user_id	INT	Owner of the document	Foreign Key → Users

Attribute	Data Type	Description	Constraints
file_name	VARCHAR(255)	Original uploaded file name	NOT NULL
file_type	VARCHAR(10)	File format (txt/pdf/docx)	NOT NULL
upload_date	DATETIME	Date and time of upload	NOT NULL
status	VARCHAR(20)	Processing status of the document	NOT NULL

0.40 Plagiarism Reports Table

Attribute	Data Type	Description	Constraints
report_id	INT	Unique identifier for the report	Primary Key, Auto Increment
document_id	INT	Document this report belongs to	Foreign Key → Documents
similarity_percentage	DECIMAL(5,2)	Overall similarity score	NOT NULL
generated_at	DATETIME	Timestamp of report generation	NOT NULL
report_file_path	VARCHAR(255)	Path to downloadable report file	NULLABLE

0.41 Similarity Results Table

Attribute	Data Type	Description	Constraints
result_id	INT	Unique identifier for a single match	Primary Key, Auto Increment
report_id	INT	Parent report	Foreign Key → Plagiarism Reports
matched_text	TEXT	The specific matched segment	NOT NULL
source_reference	VARCHAR(255)	Source of the matched content	NOT NULL

Attribute	Data Type	Description	Constraints
match_score	DECIMAL(5,2)	Similarity score for this specific segment	NOT NULL

0.42 Admin Table

Attribute	Data Type	Description	Constraints
admin_id	INT	Unique identifier for admin record	Primary Key, Auto Increment
user_id	INT	Linked user account with admin role	Foreign Key → Users
permissions	VARCHAR(255)	Comma-separated list of granted permissions	NOT NULL

0.43 Document History Table

Attribute	Data Type	Description	Constraints
history_id	INT	Unique identifier for a history entry	Primary Key, Auto Increment
document_id	INT	Related document	Foreign Key → Documents
action	VARCHAR(50)	Action performed (uploaded/checked/downloaded)	NOT NULL
action_timestamp	DATETIME	Time the action occurred	NOT NULL

8. System Constraints and Limitations

ID	Constraint / Limitation
CON-01	The current version supports only <code>.txt</code> , <code>.pdf</code> , and <code>.docx</code> file formats.
CON-02	Plagiarism detection accuracy is limited by the size and diversity of the available reference corpus.
CON-03	The system does not currently perform semantic (meaning-based) plagiarism detection; it primarily relies on textual/lexical similarity.
CON-04	The system is designed primarily for English-language documents.
CON-05	Real-time comparison against the entire public internet is not supported in the current version; comparison is limited to the internal corpus (and optionally a limited external API, if integrated).
CON-06	The system assumes a stable internet connection for both upload and report retrieval.
CON-07	Processing time increases with document size, which may affect performance for very large files.

9. Future Enhancements

The following enhancements are identified as possible directions for extending the Plagiarism Checking Portal beyond its current scope:

- **AI-assisted plagiarism detection:** Incorporating machine learning models to detect paraphrased or reworded content that simple text-matching algorithms might miss.
- **Improved semantic similarity detection:** Using vector-based embeddings to identify content that is semantically similar even when the wording differs significantly.
- **Support for multiple file formats:** Extending support to formats such as `.rtf`, `.odt`, and scanned documents (via OCR).
- **Multilingual plagiarism detection:** Extending the similarity engine to support documents written in languages other than English.
- **Improved report visualisation:** Adding interactive charts, colour-coded similarity heat-maps, and side-by-side source comparison views.
- **Integration with educational platforms:** Allowing direct submission from Learning Management Systems (e.g. Moodle, Google Classroom) via API integration.
- **Batch processing:** Allowing faculty to upload and check multiple student submissions in a single batch operation.
- **Plagiarism detection for source code:** Extending the system to detect similarity in programming assignments.

10. Acceptance Criteria

The following criteria shall be used to determine whether the developed system satisfies the requirements described in this document.

ID	Acceptance Criterion
AC-01	A new user can successfully register and subsequently log in using the registered credentials.
AC-02	A logged-in user can upload a document in a supported format and receive confirmation of successful upload.
AC-03	The system successfully extracts text from uploaded <code>.txt</code> , <code>.pdf</code> , and <code>.docx</code> files.
AC-04	The system generates a similarity percentage and a report for each successfully processed document.
AC-05	Matched segments in the report are visibly highlighted and linked to their source.
AC-06	A user can view and download a generated report without errors.
AC-07	A user's document history correctly reflects all previously submitted documents and their statuses.
AC-08	An administrator can log in to the admin panel and manage user accounts.
AC-09	The system denies access to reports and admin functionality for unauthorised users.
AC-10	The system displays a clear, user-friendly error message whenever an operation fails.

11. Conclusion

This Software Requirements Specification has presented a comprehensive description of the Plagiarism Checking Portal, covering its purpose, scope, functional and non-functional requirements, system models, database design, constraints, and planned future enhancements. The system is intended to provide students and faculty with a reliable and easy-to-use tool for detecting textual plagiarism, thereby supporting academic integrity within the institution.

The requirements outlined in this document form the basis for the subsequent design and implementation phases of the project. By clearly defining functional requirements (FR-01 through FR-43), non-functional requirements (NFR-01 through NFR-10), and acceptance criteria (AC-01 through AC-10), this SRS aims to reduce ambiguity during development and provide a clear benchmark for evaluating the completed system. It is expected that, as development progresses, this document may be revisited and refined to accommodate practical implementation considerations while remaining consistent with the core objectives described herein.

Appendix

0.44 Sample Plagiarism Report Structure

A generated plagiarism report is expected to contain, at minimum, the following sections:

- **Header:** Document title, submitted by, date of submission, date of check.
- **Summary:** Overall similarity percentage and a brief interpretation (e.g. Low / Moderate / High similarity).
- **Matched Segments Table:** A list of matched text segments, each with its corresponding source and match percentage.
- **Highlighted Document View:** The original submitted text with matched portions visually marked.
- **Footer:** System-generated disclaimer noting that the report is a supporting tool and not a final judgement of academic misconduct.

Field	Example Value
Document Title	Sample_Assignment_Report.docx
Submitted By	Student Name (Roll No.)
Date Checked	Auto-generated timestamp
Overall Similarity	e.g. 18%
Top Match Source	Reference document / corpus entry ID

0.45 Glossary

Term	Meaning
Corpus	The complete collection of reference documents used for comparison.
Similarity Threshold	The minimum matching percentage above which a segment is flagged as a potential match.

Term	Meaning
Segment	A unit of text (such as a sentence or fixed-length chunk) used as the basic unit of comparison.
Session Token	A piece of data used to identify and authorise a logged-in user's requests.
Corpus Update	The process of adding newly submitted or externally sourced documents to the reference corpus.

0.46 Requirement Traceability Matrix

Table 31 maps selected functional requirements to their corresponding use cases and system features, to support traceability throughout the development lifecycle.

Req. ID	Feature	Use Case	Verified By
FR-01 – FR-05	Registration & Login	UC-01, UC-02	AC-01
FR-06 – FR-09	Authentication & Sessions	UC-02	AC-01, AC-09
FR-10 – FR-13	Document Upload	UC-03	AC-02
FR-14 – FR-16	Text Extraction	UC-03	AC-03
FR-17 – FR-20	Plagiarism Detection	UC-03	AC-04
FR-21 – FR-23	Similarity Calculation	UC-03	AC-04
FR-24 – FR-25	Document Comparison	UC-09	AC-04
FR-26 – FR-27	Match Highlighting	UC-04	AC-05
FR-28 – FR-29	Report Generation	UC-04	AC-04, AC-05
FR-30 – FR-31	Report Viewing	UC-04	AC-06, AC-09
FR-32	Report Download	UC-05	AC-06
FR-33 – FR-34	Document History	UC-06	AC-07
FR-35 – FR-36	User Dashboard	UC-02	AC-01
FR-37 – FR-39	Admin Functionality	UC-07, UC-08	AC-08, AC-09

Req. ID	Feature	Use Case	Verified By
FR-40 – FR-41	Database Management	—	AC-04, AC-07
FR-42 – FR-43	Error Handling	—	AC-10

Table 31: Requirement Traceability Matrix